Changelog

All notable changes to RSAS (RNA Structure Analysis Suite) will be documented here.

Format based on Keep a Changelog.

[4.0.0] - 2026-11-03

Added

- **About page** (sidebar, and *RSAS > About RSAS* on macOS): version, how to cite RSAS, and the contacts (Michael M. Abdelsayed, Luiz Passalacqua) with clickable e-mail addresses.
- **RNAfold Deck** (sidebar page, `RnaThermofinder/deck/`): selects the hairpins of one or more `HP_Quality_Score_Class` tiers from RSAS Excel results, traces each to its full-length sequence, folds the full-length sequence and/or the hairpin at 1-5 temperatures and builds a PowerPoint with one verified slide per fold, plus a selection workbook, a verification CSV and per-fold JSON/SVG/PNG. Integrated from RNAfold Deck 1.3.0, with the RNAfold-webserver submission and the Chrome screenshot replaced by local ViennaRNA folding (`-p -d2 --noLP`, Turner 2004) and a local drawing (naview layout, base-pair-probability colors). No internet or browser needed; folds run in parallel processes. `tests/test_deck_local.py` checks that five saved webserver jobs are reproduced exactly (MFE, structure, per-nucleotide colors, coordinates). Each fold is also cross-checked against the RSAS sheet (MFE and structure at the same temperature). Also available as `python -m RnaThermofinder.deck run ...`.
- **Export button on the Results page** (same action as on the Analyze page), plus **Open Results Folder**.
- **One folder per analysis, named after the input**: every analysis run writes into a new folder `~/RSAS/<input>_<YYYYmmdd_HHMMSS>/`, and every file in it starts with the input file's name (extension removed): `<input>_rna_results.csv/.xlsx`, `<input>_run_log.txt` and `<input>_run_info.json` (input path, versions, temperatures, settings, file list). Earlier runs are never overwritten. RNArobo and Knotty exports go to `~/RSAS/<input>_rnarobo_<ts>/` and `~/RSAS/<input>_knotty_<ts>/` (`<input>_rnarobo_results.tsv` + `<input>_rnarobo_descriptor.des`, `<input>_knotty_results.csv`); the Sequence Extractor suggests `<genome>_extracted_sequences.fasta`. The RNAfold Deck names the deck `<input>_RNAfold_deck_<ts>`, saves it inside that analysis's folder, and prefixes every per-fold file with `<input>_`. `calculate_results_final(..., file_prefix=...)` (scripts without it keep the old names).
- **Installation for every platform** (INSTALL.md): Conda (`environment.yml` for Linux/macOS, `environment-windows.yml` for Windows), Docker (`Dockerfile`), and native venv + pip on Windows, macOS and Linux, with the optional RNArobo/Knotty engines, verification and troubleshooting. `requirements.txt` now installs ViennaRNA from its official PyPI wheels; build/test tools moved to `requirements-dev.txt`.

- **Command line** (`python -m RnaThermofinder` / `rsas-cli`): `analyze` (same pipeline and per-run output folder as the GUI), `deck` , `gui` , `version` . `pip install -e .` also installs the `rsas` GUI command.

Fixed

- Sequence Extractor: the error dialog after a failed fetch or extraction raised `NameError` (the exception variable was read by a callback after the `except` block ended).

Changed

- RNArobo searches may run for up to 48 hours (was 10 minutes), and Windows builds of RNArobo and Knotty ship in `bin/windows/` (contributed in pull requests #1 and #2).
- Minimum Python is 3.10 (the oldest version with ViennaRNA wheels).
- `python-pptx` , `matplotlib` , `pillow` and `lxml` added to the requirements; the PyInstaller build no longer excludes matplotlib and collects python-pptx's templates.

[Unreleased]

Removed

- Switch Finder is gone from the public release. The temperature-switching tool, its nav item, page, dialog, and the `switch_finder` and `switch_finder_dialog` modules, is no longer part of RSAS. Older changelog entries that mention `switch_finder.py` stay as they are, since they record what happened at the time.

Changed

- Rewrote the README, installation, and usage guides. They now cover the RNArobo Search and Pseudoknot Finder tools, the configurable RBS Window, the seven-page sidebar, and the current project layout.
- Fixed two things the old docs got wrong: the PyInstaller spec is `RSAS.spec` , not `RNAThermoFinder.spec` , and the sample file lives at `Examples/Test_Thermo_RV.fasta` .
- Bumped the package version to 3.2.0 in `setup.py` so it matches the docs, and documented the minimum Python as 3.9 to match it.

[3.2.0] - 2026-05-18

Added

- **Configurable RBS anchor and window**: the ribosome binding site search is no longer hardcoded to "5-13 nt upstream of the last AUG". A new "RBS Window" tab in Analysis Settings configures the

anchor pattern (IUPAC-aware, e.g. `DTG` matches all three bacterial start codons AUG/GUG/UUG), whether the first or last anchor match is used, and the min/max upstream spacing. Settings persist as a default or can be applied to a single run via an "Apply to this run only" checkbox, with an Analyze-screen banner when an override is active.
- `RBS_Detection_Params` **output column** (optional, default off): records the anchor/side/window used for each run, e.g. `AUG/last/5-13`.

Changed

- RBS detection is driven by a new `RbsConfig` object threaded through the analysis pipeline; the duplicated `5`/`13` window literals in `find_rbs_in_hairpin` and `find_rbs_containing_hairpin` are now a single source of truth. The AUG fallback (`find_aug_containing_hairpin`) honors the configured anchor, with its pairing threshold generalized to ceil(2/3 x L).
- Default behavior is unchanged: with default settings the RBS search is byte-identical to previous releases.

[3.1.0] - 2026-03-23

Added

- **Pseudoknot Finder (Knotty)**: new sidebar page and dialog for pseudoknot prediction using the Knotty engine (Jabbari et al., Bioinformatics 2018). Wraps the Knotty binary via subprocess, supports batch FASTA input, reports dot-bracket structure with pseudoknot brackets (`[]`, `{}`), MFE energy, and pseudoknot detection flag. Includes CSV export with formula injection sanitization, configurable max sequence length (default 500 nt), and per-sequence timeout (default 120s)
- **Switch Finder**: new sidebar page and dialog for RNA thermometer and riboswitch candidate detection via sequence truncation and temperature-dependent folding. Truncates sequence at motif boundary (upstream, downstream, or both directions), folds at each temperature with ViennaRNA, computes paired% switch score (MFE) and accessibility switch score (PF). Classifies hits as Strong/Moderate/Weak thermometer, Riboswitch candidate, Accessible, or No switch. Includes partner extraction, CSV export, and threaded batch processing

Fixed

Critical (scientifically wrong)

- **RNArobo FASTA output silently returned 0 matches**: GUI exposed a `-f` (FASTA output) checkbox, but the output parser only handles tabular format. FASTA output produced zero parsed results with no error. Removed the checkbox from the GUI
- **Riboswitch classification checked only T_low paired%**: code checked `paired_pct(T_low) >= 70%` but documentation stated "paired at all temps." Now verifies `paired_pct >= 70%` at every temperature in the series

- **Switch Finder flank parameters swapped for downstream direction**: `flank_3prime` and `flank_5prime` had opposite meaning depending on direction, producing incorrect truncation windows for downstream analysis. Renamed to direction-neutral `flank_past_motif` and `context_limit`

High

- `0.0 or -999.0` **Python falsy bug** (switch_finder.py): switch score of exactly 0.0 was treated as -999.0 in "both" direction comparison due to `or` short-circuit. Fixed with explicit `is not None` check
- **O(n^2) memory for BPP matrix** (switch_finder.py): dense `np.zeros((n,n))` replaced with `np.zeros(n)` sparse accumulation. For 10,000 nt: ~800MB → ~80KB
- **Knotty help text had malformed pseudoknot example**: `((([[..))..]]` has mismatched parentheses. Corrected to valid H-type pseudoknot notation
- **Silent T→U conversion in Knotty**: DNA sequences were converted to RNA with no user notification. Now logs conversion count when thymines are detected

Medium

- **Single-temperature switch analysis silently produced N/A**: changed validation to require >= 2 temperatures with descriptive ValueError
- **Classification threshold magic numbers duplicated**: extracted to named constants (`STRONG_THERMO_THRESHOLD`, etc.) and single `_classify_switch()` helper
- **Progress callback flooding UI**: 1000+ sequences scheduled 1000+ `after(0,...)` callbacks. Throttled to log every 10%
- **FASTA parser accepted comment lines**: lines starting with `;` were included as sequence data. Added skip for blank and comment lines
- **Knotty energy regex fallback too greedy**: standalone energy parser could match diagnostic numbers. Tightened to require number-only lines
- **Knotty IUPAC ambiguity codes silently stripped**: now warns user when N/R/Y/etc. bases are removed from sequences
- **Knotty missing simfold/params in PyInstaller build**: added `--add-data` for energy parameter files in build_app.py
- **Switch Finder sort key non-deterministic**: added `start` position as secondary sort key
- **macOS `grab_set()` timing bug in 10 dialog classes**: replaced immediate `grab_set()` with `after(100, _try_grab)` across all 8 dialog files
- **No cancel button for batch analysis**: added cancel mechanism (threading.Event + Cancel button) to Switch Finder and Knotty dialogs

Low

- **SettingsManager thread safety**: added `threading.Lock` around get/set temperatures, save_settings, and get_enabled_columns

- **Missing `openpyxl` hidden import** in PyInstaller build config
- **Knotty help text missing DP09 ligand caveat**: added note that energy model does not account for ligand-RNA interactions
- **Double T→U conversion**: FASTA parser and wrapper both converted; consolidated to wrapper only
- **CSV injection sanitization**: added `_sanitize_csv()` with quote-prefix for `=`, `+`, `-`, `@` in Knotty and Switch Finder CSV export

[3.0.1] - 2026-03-11

Added

- **RNArobo Search integration**: new sidebar page and dialog for structural motif searching using the RNArobo engine. Includes a descriptor builder GUI with preset motifs (Simple Hairpin, Two-Stem Junction, Thermometer-like), auto-fill from motif map, real-time descriptor preview, threaded search execution, and TSV export of results
- `get_user_data_dir()` helper in `settings_manager.py`: all mutable data (settings, recent files, default outputs) now stored in `~/.rsas/` so the app works correctly when distributed as a read-only macOS `.app` bundle
- macOS DMG installer with `Install RSAS.sh` script that auto-strips quarantine flags

Fixed

Critical

- `_build_pair_map` **crash on sub-sliced structures** (`HairpinAnalysis.py`): window-cut sub-structures from `find_rbs_containing_hairpin` and `find_aug_containing_hairpin` could have unmatched closing brackets, causing an `IndexError` when popping from an empty stack. Added guard: `if not stack: continue`
- **Broken** `pip install` **entry point** (`main.py` / `setup.py`): `main.py` had no `main()` function, and root-level modules (`main.py`, `settings_manager.py`) were not included in pip installs because `find_packages()` only finds directories with `__init__.py`. Added `def main()` wrapper and `py_modules=["main", "settings_manager"]` to `setup.py`

High: macOS UI freezes

- **8 messagebox calls missing** `parent=` (`quality_score_builder.py`): on macOS, messageboxes without a parent appear behind the modal dialog, freezing the entire application with no way to dismiss them. Added `parent=self.dialog` to all 8 calls
- **2 file dialog calls missing** `parent=` (`RNAGUI.py`): `filedialog.askopenfilename` and `filedialog.asksaveasfilename` could appear behind the main window. Added `parent=self.root`
- **1 confirmation dialog missing** `parent=` (`RNAGUI.py`): `messagebox.askyesno` in the clear-results handler. Added `parent=self.root`

- `quality_score_builder.py` **window close leak**: closing via the X button bypassed `grab_release()`, leaving the grab active and blocking all input. Added proper `_close()` method with `grab_release()` before `destroy()`, and `WM_DELETE_WINDOW` protocol handler

High: Data correctness

- **T→U normalization missing in** `calculate_composition` (`analysis_helpers.py`): DNA sequences (containing T) passed through without conversion, producing incorrect AU/GC/GU composition percentages. Added `.upper().replace("T", "U")`
- **Non-deterministic output sort** (`HairpinAnalysis.py`): results with tied quality scores could appear in different orders across runs, breaking reproducibility for publication. Added secondary sort key by sequence name
- **Concurrent analysis data corruption** (`RNAGUI.py`): `Cmd+R` keyboard shortcut called `run_analysis()` directly, bypassing the disabled-button check. Two concurrent analysis threads would write to the same `self.results`, progress bar, and output files. Added `_analysis_running` flag guard
- **Python 3.8 falsely advertised** (`setup.py`): code uses `list[str]` and `dict[str, ...]` syntax (PEP 585, requires Python 3.9+), but `setup.py` declared `python_requires=">=3.8"`. Updated to `">=3.9"` and removed Python 3.8 from classifiers

High: PyInstaller / distribution

- **Output directory inside read-only** `.app` **bundle** (`RNAGUI.py`): `Path(__file__).parent.parent.parent / "Data" / "Outputs"` resolves inside the `.app` on macOS, causing write failures. Now uses `get_user_data_dir() / "Data" / "Outputs"`
- **Recent files path inside read-only** `.app` **bundle** (`RNAGUI.py`): same issue with `.recent_files.json`. Now uses `get_user_data_dir() / ".recent_files.json"`
- **Settings file uses unpredictable CWD** (`settings_manager.py`): `Path("csv_output_settings.json")` resolves relative to CWD, which is unpredictable in PyInstaller. Now defaults to `get_user_data_dir() / filename`
- **Deprecated** `--deep` **codesign flag** (`build_app.py`): Apple deprecated the `--deep` flag. Removed it from the `codesign` call
- **Import ordering bug** (`HairpinAnalysis.py`): `sys.path.insert()` was placed *after* `from settings_manager import SettingsManager`, so the module could only be imported if something else had already modified `sys.path`. Moved path setup before all project imports
- **Missing** `sys.path` **setup** (`settings_dialog.py`): imported `from settings_manager import DEFAULT_TEMPERATURES` without adding the project root to `sys.path`. Only worked by accident when imported through `RNAGUI.py`. Added explicit path setup

Medium

- **3 silent partition function exceptions** (`HairpinAnalysis.py`): `except Exception: pass` blocks at PF computation sites silently swallowed errors, making failures invisible. Changed to log warnings to stderr

- **Silent motif analysis exception** (`HairpinAnalysis.py`): broad `except Exception: pass` hid failures in motif analysis. Changed to log with sequence name to stderr
- **Mutable default arguments** (`HairpinAnalysis.py`): `temps=[25,37,42]` and `temps: List[int] = [25, 37, 42]`, mutable defaults shared across calls. Changed to `temps=(25,37,42)` and `Optional[List[int]] = None` with guard
- **Empty motif guard** (`motif_finder.py`): empty or whitespace-only motif pattern caused unnecessary processing. Added early return for empty motifs
- **Fractional temperatures silently truncated** (`settings_dialog.py`): float values like `37.5` were silently cast to `int`. Now explicitly rejected with an error message
- **IntVar crash on CPU cores** (`settings_dialog.py`): non-integer input in the CPU cores field raised `TclError`. Wrapped in try/except with user-friendly error message
- **Tooltip dark mode** (`settings_dialog_csv.py`): hardcoded light-mode colors (`#f8f9fa` background) appeared as bright white boxes in dark mode. Now detects appearance mode and uses appropriate colors
- **Tooltip Toplevel leak** (`settings_dialog_csv.py`): rapidly hovering between elements could create multiple tooltip windows. Added `_hide()` cleanup before creating new tooltip, with `TclError` guard on destroy
- **KeyError on corrupted settings** (`settings_dialog_csv.py`): direct dict access `self.settings_manager.settings["csv_output_columns"]` could crash on corrupted settings files. Changed to `.get("csv_output_columns", {})`
- **Toast race condition** (`settings_dialog_csv.py`, `sequence_settings_dialog.py`): toast duration (3500ms) exceeded dialog destroy timing (1200ms), causing `TclError` on dismiss. Reduced to `duration=1000`
- **Drag-and-drop path corruption** (`RNAGUI.py`): `strip("{}")` would strip any of those characters from anywhere in the path. Changed to proper brace-pair stripping: `if path.startswith("{") and path.endswith("}"): path = path[1:-1]`
- `_open_recent` **ValueError** (`RNAGUI.py`): `self._recent_files.remove(path)` could raise `ValueError` if path was already removed. Added existence check before removal
- `.as_posix()` **in sys.path** (`settings_dialog_csv.py`): used `.as_posix()` which produces forward-slash paths on Windows. Changed to standard `str(Path(...))`

Low

- **Toast** `_dismiss` **exception scope too broad** (`RNAGUI.py`, `settings_dialog_csv.py`, `sequence_settings_dialog.py`): `except Exception` in toast dismiss handlers caught everything; narrowed to `except tk.TclError`
- **Unused import** (`RNAGUI.py`): removed unused `SettingsDialogModern` import
- **JSON I/O encoding** (`settings_manager.py`): missing `encoding='utf-8'` on file open calls, could cause encoding issues on some systems. Added explicit encoding
- **Save error handling** (`settings_manager.py`): only caught `IOError`; added `TypeError` and `ValueError` for robustness

Known Issues (deferred, require design decisions)

- `quality_scoring.py`: weight clamping at `max(1.0)`, needs a decision on whether to allow fractional weights
- `quality_scoring.py`: N/A metrics count as 0 in the denominator, needs a decision on scoring methodology
- `settings_manager.py`: `set_temperatures()` does not auto-update scoring profiles, needs a migration strategy
- `sys.path` hacks across 5 files: functional but inelegant; the proper fix is moving `settings_manager.py` into the package

[3.0.0] - 2025-02-23

Added

- Rebranded from RNA Thermometer Finder to RSAS: RNA Structure Analysis Suite
- Modern CustomTkinter GUI with sidebar navigation, dark/light mode support
- Dual quality scoring system (hairpin 0-6, full-length 0-6)
- Drag-and-drop file input
- Keyboard shortcuts (Cmd/Ctrl+O, R, E)
- Toast notifications for user feedback
- **Customizable folding temperatures**: configure 1-5 folding temperatures instead of the fixed 25/37/42. All output columns, CSV headers, and Excel tabs update dynamically based on the configured temperatures
- **Motif / Sequence Finder**: search for any user-defined nucleotide pattern (IUPAC-aware, including degenerate codes like R, Y, N) across all input sequences. Reports paired percentage, dot-bracket structure, and partition-function accessibility at each temperature. Includes temperature difference columns to detect thermometer-like behavior
- **All motif matches reported**: every overlapping occurrence of the motif is shown (semicolon-separated in the CSV summary columns). A dedicated "Motif Matches" Excel tab provides one row per hit per sequence with full detail
- **Synthetic Pool Generator**: generate pools of random RNA sequences with user-defined segment layouts (random regions + fixed IUPAC motifs). Includes a preset for RBS+AUG (R84 + GGAGG + R8 + AUG). Optional composition filtering by GC%, AU%, and/or GU% with independent target and tolerance per metric. Output is FASTA format
- **Terminal Hairpin Quality Score Builder**: interactive dialog for configuring the criteria and weights used in the hairpin quality scoring system
- **Bidirectional upstream sequence extractor**: supports extracting sequences from either side (upstream or downstream) of a feature, with configurable inclusion/exclusion of the start codon
- **Separate settings dialogs**: Analysis Settings (hairpin detection + folding temperatures) and Performance Settings (CPU cores) are now independent cards with their own dialogs

- **Sidebar navigation for new tools**: Synthetic Pool Generator and Sequence Extractor each have their own page accessible from the sidebar

Changed

- Complete GUI rewrite from legacy tkinter to CustomTkinter
- Card-based settings page with separate Analysis Settings and Performance cards
- Excel export now supports up to 3 tabs (Full Sequence, Hairpin Analysis, Motif Matches) depending on enabled features
- All internal column key generation uses the configured temperature list dynamically

Fixed

- `gc_content()` in HairpinAnalysis.py now handles empty strings without ZeroDivisionError
- Synthetic pool dialog segment type callback correctly captures the segment index
- Consistent use of `t_first` variable in hairpin Excel column builder

[2.1.0] - 2025-02-14

Added

- RBS-based hairpin detection with AUG fallback for fourU-type thermometers
- Upstream sequence extractor (local files and NCBI fetch)
- Sequence Options dialog for appending sequences before analysis
- Partition function columns (ensemble energy, mean paired prob, RBS accessibility)
- Full-length RBS sequestering at each temperature with difference columns
- Output Columns presets (Hairpin, Full Sequence, Riboswitch, Full Export) and custom presets
- Excel export with Full Sequence and Hairpin Analysis tabs
- Tooltips throughout the GUI

Changed

- CPU cores default to 1 (sequential) instead of auto
- Export flow uses Save As with timestamped filename suggestions

[2.0.0] - 2024-12-18

Added

- Original sequence quality scoring (0-6)
- Original sequence MFE and composition range filters

- Conditional calculation system (only computes enabled columns)

Changed

- Analysis Settings includes original sequence ranges
- Quality score naming: Quality_Score to Quality_Score_Hairpin

Fixed

- Fallback CSV headers include all original sequence columns
- Column mapping consistency in analysis_helpers.py

[1.0.0] - 2024-11-10

Added

- Initial release
- Terminal hairpin detection
- Temperature-dependent MFE analysis (25, 37, 42)
- Base pair composition analysis (AU%, GC%, GU%)
- RBS identification and quality scoring (0-6)
- GUI interface with CSV export
- macOS .app bundle distribution